

KALAOA KONA, HI, USA

WMO: 911975

Lat: 19.736N

Lon: 156.049W

Elev: 13

StdP: 101.17

Time Zone: -10.00 (W10)

Period: 97-19

WBAN: 21510

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.9	19.8	10.8	8.0	22.3	12.1	8.8	22.4	13.2	25.6	11.7	26.0	2.7	70	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	31.2	24.5	30.6	24.1	30.0	23.8	25.5	29.8	25.0	29.2	24.6	28.8	5.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.0	28.5	23.5	18.4	28.2	23.0	17.7	27.9	77.9	29.7	76.0	29.3	74.4	28.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	7.5	6.6	DB	17.0	32.6	1.0	2.0	16.3	34.0	15.7	35.1	15.2	36.2	14.5	37.6
			WB	12.6	25.6	1.6	1.0	11.5	26.4	10.5	27.0	9.6	27.6	8.5	28.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	23.8	23.7	24.2	25.0	25.5	26.2	26.7	27.2	27.0	26.6	25.5	24.4
	DBStd	1.53	0.91	1.00	0.98	0.89	0.85	0.83	0.96	1.03	0.99	0.94	0.89	0.90
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5658	427	384	442	449	482	485	518	534	511	514	466	446
	CDD18.3	2616	169	151	183	199	223	235	260	276	261	256	216	187
	CDH23.3	21839	939	828	1085	1384	1779	2157	2607	2909	2718	2520	1743	1170
	CDH26.7	4369	55	52	96	169	269	427	641	831	768	645	305	112
Wind	WSAvg	3.5	3.5	3.6	3.6	3.6	3.5	3.5	3.5	3.5	3.3	3.2	3.3	
Precipitation	PrecAvg	715	66	37	57	53	53	69	71	71	74	64	52	49
	PrecMax	1383	182	95	155	267	126	143	177	154	150	155	155	196
	PrecMin	426	0	0	11	23	10	15	24	37	21	7	8	0
	PrecStd	233	50	27	37	49	28	34	36	27	31	37	41	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	28.6	29.0	29.3	29.7	30.5	31.2	32.0	31.9	31.3	30.1	29.0
		MCWB	20.7	21.8	21.1	22.1	23.0	23.3	24.6	25.3	25.2	24.6	22.8	21.3
	2%	DB	27.7	27.8	28.1	28.7	29.1	29.7	30.3	31.1	31.1	30.5	29.2	28.1
		MCWB	21.1	21.5	21.7	22.2	22.8	23.1	24.0	24.7	24.7	24.0	22.8	21.6
	5%	DB	27.3	27.4	27.6	28.1	28.6	29.1	29.8	30.3	30.2	29.9	28.8	27.7
		MCWB	21.1	21.3	21.6	21.9	22.7	22.9	23.8	24.2	24.1	23.7	22.7	21.6
	10%	DB	27.0	26.9	27.2	27.7	28.0	28.7	29.1	29.8	29.7	29.2	28.2	27.4
		MCWB	20.9	21.0	21.4	21.7	22.3	22.8	23.4	23.9	23.7	23.4	22.5	21.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.6	24.0	23.9	24.5	24.6	25.4	26.0	26.1	25.7	24.6	23.9
		MCDB	26.8	27.0	27.2	27.8	28.0	28.9	29.7	30.5	30.6	29.9	28.4	27.1
	2%	WB	22.9	22.9	23.2	23.2	23.8	24.1	24.8	25.4	25.4	24.9	24.1	23.1
		MCDB	26.3	26.8	26.7	27.3	27.6	28.2	29.1	29.7	29.7	29.1	28.0	26.7
	5%	WB	22.2	22.3	22.6	22.8	23.3	23.8	24.4	25.0	24.9	24.5	23.7	22.7
		MCDB	26.1	26.3	26.5	27.1	27.4	28.0	28.7	29.2	29.2	28.7	27.7	26.5
	10%	WB	21.7	21.7	21.9	22.3	22.9	23.3	24.0	24.5	24.4	24.0	23.1	22.2
		MCDB	25.8	25.8	26.3	26.8	27.1	27.7	28.3	28.8	28.7	28.3	27.3	26.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.6	6.2	5.9	5.6	5.6	5.6	5.8	5.9	6.0	6.1	6.3
		MCDBR	6.8	6.8	6.6	6.2	6.0	5.9	6.0	6.1	6.3	6.3	6.4	6.7
		MCWBR	4.2	4.2	3.9	3.7	3.5	3.6	3.5	3.6	3.8	3.8	3.9	4.2
	5% WB	MCDBR	6.3	6.1	6.1	5.9	5.7	5.7	5.7	5.7	5.9	5.9	5.8	5.9
		MCWBR	4.2	4.1	3.9	3.6	3.4	3.5	3.4	3.5	3.7	3.8	3.9	4.0
Clear-Sky Solar Irradiance	taub	0.361	0.367	0.383	0.423	0.412	0.390	0.399	0.385	0.375	0.379	0.375	0.357	
	taud	2.582	2.565	2.518	2.374	2.443	2.508	2.486	2.536	2.569	2.575	2.581	2.614	
	Ebn at Noon	920	932	925	885	884	896	888	907	919	908	898	913	
	Edh at Noon	93	99	108	125	116	108	110	105	101	97	92	88	
All-Sky Solar Radiation	RadAvg	4.62	5.08	5.53	6.14	6.40	6.58	6.67	6.53	6.07	5.29	4.56	4.22	
	RadStd	0.23	0.32	0.56	0.40	0.38	0.21	0.23	0.29	0.20	0.19	0.19	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+125

Nomenclature: See separate page